Trial SET #5 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only 60 minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

Name of the student _____

1. In an art gallery, 25 percent of the art pieces on display were photographs. If 12 percent of the photographs were black-and-white photographs, what percent of the art pieces on display were black-and-white photographs?

A. 3% B. 7% C. 13% D. 30% E. 37%

2. If A and B are positive integers, which of the following expressions is not an integer for certain?

A. $(2A^2 - 2B^2)/(A+B)$; B. $(6B + 8A)/(3B + 4A)$; C. $(3A - B)/(B - 3A)$.;
D. $(A + B)/(A^2 + B^2 + 2AB)$.; E. $(A^2-B^2)/(A-B)$.;

3. If $X \sim Y = X^Y - Y^2$, then what is the value of $(3) \sim(-2)$?
(a) 5
(b) 14
(c) -35/9
(d) 37/9
(e) 2

4. A certain taxi fare consists of an initial charge of $1.25 and an additional charge of $0.25 for each 1/5 mile traveled. What is the total fare for a trip of 2.4 miles?

(A) $4.25 (B) $3.00 (C) $2.25 (D) $1.85 (E) $1.75

5. If k is a positive integer, which of the following is equivalent to $\left(2k^{\frac{1}{2}}\right)^{-2}$

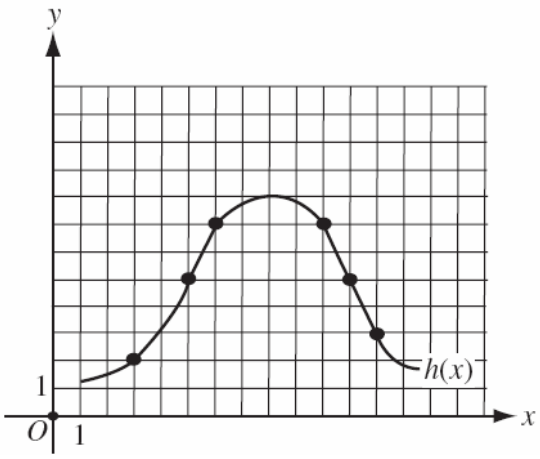
A. $\frac{2}{k}$ B. $\frac{1}{2k}$ C. $\frac{1}{4k}$ D. $\frac{1}{2k^2}$ E. $\frac{4}{k}$

6. R, S, T, and U are points on a line, and U is the midpoint of line segment ST. If the lengths of line segments RS, RT, and ST are 20, 4, and 24, respectively. What is the length of line segment RU?

A. 6 B. 8 C. 12 D. 14 E. 16

7. How many different 6-letter sequences are there that consist of 1 A, 2 B's, and 3 C's ?

A. 6 B. 60 C. 120 D. 360 E. 720



8. In the figure above, if the value of $h(5) = k$, then what is the value of $h(2k)$?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

$(x-8)(x-k)=x^2-5kx+m$

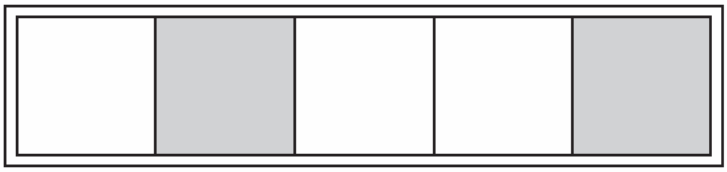
9. In the equation above, k and m are constants. If the equation is true for all values of x, what is the value of m?

(A) 8
(B) 16
(C) 24
(D) 32
(E) 40

10. Given that $f(x)=\frac{x-3}{2}$, $f^{-1}(x)+f^{-1}(2x)=12 \Rightarrow x=?$

($f^{-1}(x)$ represents the inverse fumction)

A. 2; B. 1; C. 1.5; D. 0.5; E. 2.5;



11. The figure above shows 5 cubbyholes in a day care center. Five different-colored jackets—one red, one blue, one green, one brown, and one white—are to be randomly placed in these cubby holes, one jacket per cubby hole. What is the probability that the red and brown jackets will each be assigned to one of the cubbyholes indicated by the shaded squares?

A. 1/4; B. 1/6; C. 1/10; D. 0.05; E. 1/40;

12. At a certain company, each employee has a salary grade s that is at least 1 and at most 5. Each employee receives an hourly wage p, in dollars, determined by the formula $p = 9.50 + 0.25(s - 1)$. An employee with a salary grade of 5 receives how many more dollars per hour than an employee with a salary grade of 1?

A. \$0.50 B. \$1.00 C. \$1.25 D. \$1.50 E. \$1.75

13. If $2\sin^2 x + \sin x \cdot \cos x = 1$, then $\tan 2x = ?$

A) 4 B) 2 C) 1 D) $\frac{1}{2}$ E) $\frac{1}{4}$

14. Find the simplest firm of the following expression $1-8\sin^2\beta\cos^2\beta$.

A) $\cos 4\beta$ B) $\sin^2\beta$ C) $\cos^2\beta$ D) $\sin 4\beta$ E) $\sin\beta$

15. The numbers x and y are three-digit positive integers, and $x + y$ is a four-digit integer. The tens digit of x equals 7 and the tens digit of y equals 5. If $x < y$, which of the following must be true?

I. The units digit of $x + y$ is greater than the units digit of either x or y.
II. The tens digit of $x + y$ equals 2.
III. The hundreds digit of y is at least 5.

A. II only
B. III only
C. I and II
D. I and III
E. II and III

16. If s, u, and v are positive integers and $2^s = 2^u+2^v$, which of the following must be true?

I. $s = u$
II. $u \neq v$
III. $s > v$

(A) None (B) I only (C) II only (D) III only (E) II and III

17. In the coordinate plane, a circle has center (2, -3) and passes through the point (5, 0). What is the area of the circle?

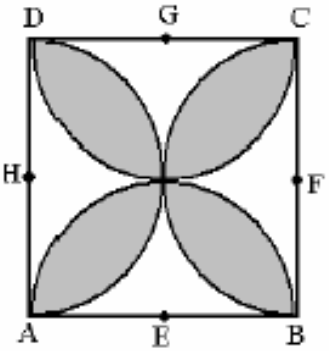
A. 3π B. $3\sqrt{2}\pi$ C. $3\sqrt{3}\pi$ D. 9π E. 18π

18. After the fist term in a sequence, the ratio of each term to the preceding term is x. If the first term is y, what is the third term?

A. $\frac{y}{x}$; B. x^2y^2; C. $\frac{x}{y}$; D. $\frac{x^2}{y}$; E. x^2y;

19. In the youth summer village there are 150 people, 75 of them are not working, 50 of them have families and 100 of them like to sing in the shower. What is the largest possible number of people in the village, which are working, that doesn't have families and that are singing in the shower?

a) 25.
b) 50.
c) 75.
d) 100.
e) 150.



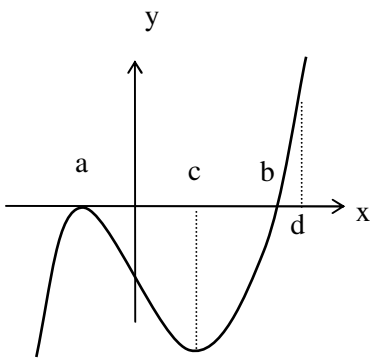
20. ABCD is a square with side 1 cm. and E, F, G and H are centers of given semicircles. Find the shaded area.

A. $\frac{\pi}{2}-1$; B. $\pi-1$; C. $\frac{\pi}{3}-1$; D. $\frac{\pi}{4}-\frac{1}{4}$; E. $2\pi-6$;

21. $\log_3 5 = m$ and $3^{x+4} = 5^{4x+1} \Rightarrow$ x = ?

A) $\frac{m}{1-m}$ B) $\frac{m-4}{4m-1}$ C) $\frac{m-4}{4m+1}$ D) $\frac{4-m}{4m-1}$ E) $\frac{4-m}{1-4m}$

22.



Which one of the followings is false for the function f given in the figure?

A) $f'(0) < 0$ B) $f(d) > 0$ C) $f'(a) = 0$
D) $f''(c) = 0$ E) $f'(b) > 0$

23. Find the area of the region bounded by the curve $y = 2 - \frac{x^2}{2}$ and x axis.

A) 14/3 B) 16/3 C) 17/3 D) 6 E) 7

24. A, B, C are three consecutive positive integers (A > B > C). What is the value of the expression 2A + B + 3C?

A. 6A+7; B. 5A+1; C. 5A-1; D. 6A-5; E. 6A-7;

25. A grocery store ordered a delivery of fresh milk products that contained 45 milk bottles, 24 cheese packs and 23 cartons of chocolate milk. If the chocolate milk carton costs like a bottle of milk, which is three times the price of a cheese pack, what fraction of the cost of the entire purchase was the cost of 20 bottles of milk, 1 pack of cheese and 5 chocolate milk cartons?

A. 1/6; B. 2/5; C. 1/4; D. 1/3; E. 3/7;

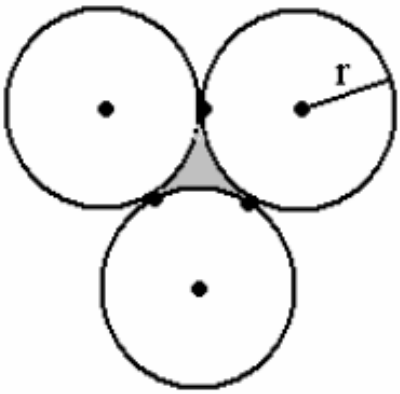
26. $\int (\frac{2x+3}{2x+1})dx = ?$

A. $\frac{4x-6}{(2x+1)^2}+C$; B. $\frac{8x+8}{(2x+1)^2}+C$; C. x + ln(2x+1) $+C$;

D. x + 2ln(2x+1) $+C$; E. $\frac{2}{(2x+1)^2}+C$;

27. Find the equation whose is tangent to the $f(x) = 2x^2-1$ is at $x_0 = 0$.

A) f(x) = 3x B) f(x) = -1 C) f(x) = 2
D) f(x) = x+1 E) f(x) = 1-x



28. In the given figure the three circles with radii ***r*** are externally tangent to each other. If the shaded area is $\sqrt{3}-\frac{\pi}{2}$ then find ***r.***

A) 1 B) 2 C) 3 D) 4 E) 5

29. There is a toy house which is a small copy of a true existing house. Little Jimmy measures an area of entrance door and conclude that it is 64 times less than actual area. If the volume of Jimmy's room is $64m^3$ then find the volume in m^3 of the corresponding room in the toy house.

A) 1 B) 0.5 C) 0.125 D) 4 E) 0.0125

30. Seven members of the school band—Aretha, Benny. Charles, Darryl. Ella, Frances and Gerald— have been selected to play a special jazz tribute for the governor's office. For the tribute, the governor's office will arrange these members standing in a row of seven spots on a platform, subject to the following restrictions:
Charles must stand in the middle spot.
Aretha must stand in the left-most spot.
There must be exactly two spots between Benny and Frances.
Darryl cannot stand next to Charles.
In which of the following pairs could <u>neither</u> person be placed in the last position from the left?

A. Benny and Darryl
B. Darryl and Aretha
C. Charles and Ella
D. Benny and Frances
E. Ella and Gerald